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Enhancing anemia detection through multimodal data fusion: a non-invasive approach using EHRs and conjunctiva images

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Abstract
Anemia detection using multimodal approaches leverages the integration of multiple data sources, such as imaging, clinical records, and hematological parameters, to improve diagnostic accuracy. Such methods can capture the complex interplay of factors contributing to anemia, providing a more comprehensive assessment than traditional single-modality techniques. In this research, a novel deep learning multi-modal feature fusion approach is proposed for the automated detection of anemia using EHRs (Electronic Health Records), and Conjunctiva image dataset. First, EHR records are preporcessed by selecting the most appropriate features using Random Forest. The features from the conjunctiva images are extracted using RCBAM (Reverse Convolution Block Attention Mechanism). After that, GRAD-Cam algorithm is applied to calculate the pixel percentages of all the features. The output of Random Forest and GRAD-Cam algorithms is concatenated to form a multimodal fusion. The important information from the concatenated features is selected with the help of a professional healthcare consultant. The different experiments are performed on textual and image datasets individually and after concatenating. The results show that the proposed model outperforms from state-of-the-art methods with an accuracy of 95%. Despite challenges such as class imbalance and computational demands, our findings reveal substantial clinical potential, offering a patient-friendly and accessible diagnostic solution.

Keywords Multimodal · Non-invasive technique · Anemia detection · Handcrafted features · RCBAM

Abbreviations

HB	Hemoglobin
RBCs	Red blood cells
WHO	World Health Organization
CNN	Convolutional neural network
LR	Logistic Regression
SVM	Support vector machine
k-NN	K-nearest neighbor
SMOTE	Synthetic minority over-sampling technique
ADASYN	Adaptive synthetic sampling approach

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RGB	Red green blue
NB	Naïve bayes
RF	Random forest
HER	Electronic health records
RCBAM	Reverse_Convolutional Block Attention Module
CBAM	Convolutional block attention module
ROC	Receiver operating characteristics
AUC	Area under the curve

1 Introduction

In the realm of global health, anemia emerges as a significant concern affecting people worldwide, with approximately 25% of the global population impacted, as reported by the World Health Organization (WHO) [1]. The urgency to address anemia stems from its profound consequences across various health aspects, hindering cognitive development in children, complicating pregnancy, and increasing susceptibility to infections. Recognizing anemia as a critical health challenge necessitates exploration beyond conventional diagnostic methods [2]. Robust diagnostic measures are imperative to mitigate its extensive impact and enable effective intervention. Early and accurate detection is crucial for preventing adverse health outcomes, as timely identification allows prompt intervention, reducing the risk of complications. Typically, hemoglobin (Hb) concentration in blood is less than 13.5 g/dl in adult males and less than 11.5 g/dl in females, no discernible symptoms manifest, as the body strives to compensate for oxygen loss. For instance, it increases heart rate in response to low oxygen levels. However, when compensatory mechanisms can no longer ensure sufficient oxygen levels, the symptoms above become evident.

According to the WHO [3], achieving a comprehensive and reliable assessment of anemia on a global scale, as well as determining its full scope and impact, poses significant challenges, even in more economically developed regions. What is undeniable, however, is that anemia exerts a substantial toll on the world's population, with profound implications for both social and economic development. All countries grapple with the financial burdens associated with addressing this condition, including prevention efforts (or the absence thereof), the provision of traditional diagnostic tools, administrative costs, and healthcare accommodations. Iron deficiency anemia is known to impact cognitive functions and physical growth in children, as well as reduce productivity in adults [4]. The risk of developing anemia is also heightened in individuals with chronic illnesses. Various medical conditions are associated with a higher prevalence of anemia, including diabetes, kidney disorders, cancer, human immunodeficiency virus (HIV/AIDS), inflammatory bowel disease (IBD), and cardiovascular diseases [5]. Additionally, diseases such as malaria, schistosomiasis (bilharzia), and Hb disorders are significant contributors to the occurrence of anemia [5, 6].

However, current challenges in anemia detection include accuracy and speed limitations of conventional methods, variations in Hb levels, and the discomfort associated with invasive techniques like blood tests [7, 8]. Non-invasive diagnostic techniques are essential to overcome these limitations, improving patient compliance and enabling frequent monitoring. Multimodal and feature fusion [9] offers a promising solution by integrating diverse information sources, including blood-related parameters, imaging data, and clinical data. This approach provides a more nuanced understanding of anemia, facilitating a sophisticated diagnostic process. Research focusing on developing intelligent, non-invasive techniques for anemia detection through multimodal data fusion aligns with the evolving landscape of medical diagnostics, promising significant advancements in precision and effectiveness.

Blood tests [10], particularly those assessing Hb levels and red blood cell count, have long been fundamental in anemia detection. These tests directly measure Hb, offering insights into the blood's oxygen-carrying capacity. However, their interpretation is influenced by various factors such as age, sex, and altitude, leading to result variability. Hb levels naturally vary with age and differ between genders, while individuals at higher altitudes may exhibit elevated levels due to adaptation to lower oxygen levels [11]. On the other hand, microscopic examination of blood smears allows for a detailed analysis of red blood cells (RBCs), aiding in the detection of abnormalities in size, shape, and color. Despite its valuable insights, this method is labor-intensive, time-consuming, and requires specialized expertise, introducing potential subjectivity and interobserver variability in interpretation [12].

In recent years, there has been a growing interest among researchers in exploring innovative methods and tools for monitoring Hb (Hb) concentrations noninvasively and affordably, suitable for both laboratory and at-home applications. While doctors can suspect anemia through visual examination of indicators like the palms of the hands, fingertips, nail

beds, and eye conjunctiva, a definitive diagnosis is often elusive. Hence, the development of noninvasive, cost-effective, and easily accessible devices and techniques for diagnosing anemia through imaging these bodily areas are crucial. Machine Learning (ML) techniques have gained widespread and effective application across various fields, particularly in the realm of medicine, including the assessment of anemia.

2 Literature review

Recent advancements in computational techniques have introduced promising approaches to overcome these limitations. Computer vision techniques [13] analyze facial and conjunctival images to detect anemia, utilizing visual cues such as pale skin and scleral icterus. However, these methods lack specificity and may not capture subtle variations. Retinal imaging offers a non-invasive alternative, examining blood vessels at the back of the eye to identify changes indicative of anemia, such as increased central retinal vein pressure. Yet, this approach relies on specialized equipment and may not be widely accessible [14]. Traditional methods for anemia detection have several limitations that impede their effectiveness. Firstly, there is variability [2] in results, particularly in numerical values like Hb levels, influenced by demographic factors and environmental conditions. Additionally, the invasiveness of blood tests [2], involving needles, can deter individuals from undergoing regular screenings due to discomfort or fear. Moreover, clinical symptoms, while indicative, are subjective and may not definitively determine the presence or severity of anemia [15].

The RGB components [16] are extracted and utilized the Naïve Bayes algorithm which achieved an accuracy of 92.3%. In [17], the CNN model achieved the highest accuracy for anemia detection. However, when this model was integrated into a FastAPI server as demonstrated in [18], where a conjunctiva detection model and anemia detection model were developed using a combination of Convolutional Neural Networks (CNN), Logistic Regression (LR), and the Gaussian Blur algorithm, the overall accuracy was impacted. These models were deployed to a FastAPI server and connected to a mobile application built using React Native, which further affected the model's performance when utilized on mobile devices.

In addition, the study [19] compared the results with those results which are obtained from the laboratories with a specificity and sensitivity of 85.18% and 91.89% similarly, in the context of conjunctiva and tongue images, when both LR and neural network algorithms were applied. Numerous research endeavors have been carried out to develop a non-invasive method for detecting anemia, particularly by utilizing conjunctiva images of the eye in [20], and In [21] Support Vector Machine (SVM) algorithms were employed to automate the creation of a non-invasive model for anemia detection using 19 conjunctival images. These images were associated with different known Hb levels, achieving an accuracy of 78.90% in 15 out of the 19 cases. Additionally [21], utilized the least squares SVM (LS-SVM) in conjunction with image processing and computer vision techniques to identify anemia from conjunctiva images. This approach yielded a precision rate of 85%, a sensitivity of 92%, and a specificity of 70%. Their study involved 77 tested images, of which 21 were from non-anemic individuals, and 56 were from anemic individuals.

Furthermore, anemia detection has been explored through conjunctival images, where a novel device was introduced for image capture, primarily employing the k-Nearest Neighbor (k-NN) algorithm. This approach demonstrated significant performance, particularly when tested with non-anemic images across 113 instances, proving its efficacy [4]. Similarly, in the context of anemia detection, Decision Tree (DT), SVM, and K-NN algorithms were harnessed by [22], utilizing pallor digital images. Among these models, the DT exhibited the highest accuracy, reaching 82.61% in performance compared to the others. Furthermore, [23] employed a CNN in their research for anemia detection, achieving an impressive accuracy of 94% when utilizing the Adam optimizer as show in Table 1.

In the realm of medical image analysis, several studies have explored techniques for segmenting and classifying ocular and oral tissues. One study [20] focused on segmenting 19 conjunctival eye images using an RGB thresholding algorithm, achieving an accuracy of 78.90%. In a separate study [24], the emphasis shifted to the conjunctiva, where image augmentation and segmentation, coupled with manual extraction of the region of interest, led to an impressive 97.00% accuracy using artificial neural networks (ANN). Lastly, an investigation [25] delved into both conjunctival and tongue images, employing diverse feature extraction methods and classifiers. Notably, the DT algorithm yielded an 86% accuracy for conjunctival images and an impressive 98.2% accuracy for palpable tongue images. This comparative analysis showcases the diverse methodologies employed in prior works, offering a comprehensive perspective on the advancements in image analysis for ocular and oral tissues.

While existing methods have contributed to anemia detection, they often suffer from invasiveness, limited accessibility, or lack of specificity. The proposed technique, leveraging multimodal data fusion, aims to address these challenges, providing a promising avenue for advancing anemia detection methodologies. Comparative studies will be crucial to

Table 1 The comparison of the state-of-the art methods for anemia detection

Ref	Dataset	Number of images	Algorithms	Accuracy
[16]	Self-Collected	40 Respondents Images	Naïve Bayes	92.3%
[18]	CP-AnemiC (A Conjunctival Pallor) Dataset from Ghana	1844 Train, validate 264 and 527 test conjunctiva Images	CNNs (CNN), LR, and Gaussian Blur algorithm	92.50%
[21]	Self-Collected	19 Conjunctival Images	Least Square SVM(LS-SVM)	85%
[22]	104 Patients Images	84 Train data 23 Test data	DT, SVM, KNN	82.61%
[23]	2000 Images	1440 Train, 160 Validation, and 400 Test Images	CNN	94%

evaluate the performance of the proposed technique against established approaches, ensuring its efficacy and potential integration into clinical practice.

3 Methodology

In response to the limitations of existing methods, this research proposes an intelligent, non-invasive technique for anemia detection using multimodal data fusion, which integrates information from Electronic Health Records (EHRs), and conjunctiva images. This approach incorporates a novel module, the Reverse Convolutional Block Attention Module (RCBAM), into a modified MobileNetV2 architecture. This enhanced feature extraction method for conjunctival images has not been previously used in anemia detection studies. By combining data from diverse sources, the proposed technique overcomes the shortcomings of individual modalities, achieving superior diagnostic accuracy and precision. Notably, the model attained the highest accuracy and precision compared to previous models related to conjunctiva studies for anemia. The key advantages of this technique include non-invasiveness, eliminating the need for invasive procedures, and ensuring a patient-friendly and accessible diagnostic process. Moreover, multimodal data fusion [26] enhances the accuracy and reliability of anemia detection compared to single-modality methods, potentially enabling early-stage detection and timely intervention. Additionally, the technique offers a more personalized diagnosis by considering individual patient characteristics and risk factors through the fusion of multimodal data.

The experiments were conducted under the oversight of a healthcare professional, specifically a doctor with three years of field experience and holding an MBBS degree from Shandong First Medical University, China, who was practicing at Sahiwal Medical College (SLMC). All methods were performed in accordance with the relevant guidelines and regulations. The experimental protocol received approval from the overseeing doctor. Additionally, as the dataset utilized in this study is publicly available online, consent from all subjects was inferred from the publicly accessible nature of the data.

3.1 Overview

This study presents a five-step process for implementing a non-invasive model for anemia detection. The first step provides an overview of the model's development and objectives which includes the data collection and preprocessing. The second step involves feature extraction, where relevant data from Electronic Health Records (EHR) and conjunctiva images are identified and extracted. In the third step, these features are combined using multimodal fusion algorithms to create a unified dataset. The fourth step focuses on the classification, where a machine learning model is trained and evaluated using the fused features. Finally, the fifth step outputs anemia predictions based on the trained model. This approach leverages multimodal data and advanced machine learning techniques to offer a comprehensive and non-invasive solution for anemia detection as shown in Fig. 1.

3.2 Data collection

- Textual dataset

  The methodology employed in this research draws upon two principal data sources. Firstly, EHRs furnish a wealth of clinical information encompassing patient demographics, medical histories, and laboratory findings [27]. This data repository offers crucial insights into overall health status and potential factors contributing to anemia risk.



Fig. 1 The overview of the proposed approach for anemia detection

- Image dataset

  The dataset, collected by trained biomedical scientists at the Centre for Research in Applied Biology, University of Energy and Natural Resources, Ghana [27], comprises conjunctival images from children aged 6–59 months across 10 hospitals. Captured using Kobo Collect v2021.2.4 on Samsung Galaxy Tab 7A tablets, it includes patient biodata and remarks on Hb values. Images were taken under ambient natural light, with camera spotlights off to minimize shine effects. The region of interest within the conjunctiva was extracted using the triangle thresholding and entropy grayscale algorithms, enhancing its utility for biomedical research.

3.3 Evaluation metrics

3.3.1 Accuracy

Accuracy represents the proportion of correctly predicted instances (both true positives and true negatives) out of all instances, providing a general measure of how well the model performs across all classes. The mathematical equation for accuracy is

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

3.3.2 Area under the curve

The AUC represents the degree or measure of separability between classes, indicating how well the model distinguishes between positive and negative instances. A higher AUC score means the model is better at predicting both true positives and true negatives. It is calculated as

$$AUC = \int_0^1 \frac{TP}{TP + FP} d\left(\frac{FP}{FP + TN}\right)$$

3.3.3 Precision

Precision measures the proportion of correctly predicted positive instances among all instances predicted to be positive, focusing on the quality of positive predictions. This measure can be calculated as

$$Precision = \frac{TP}{(TP + FP)}$$

3.3.4 Recall

Recall, also known as sensitivity or the true positive rate, measures the proportion of correctly predicted positive instances out of all actual positive instances, representing the model's ability to identify positive instances. It is calculated as:

$$Recall = \frac{TP}{(TP + FN)}$$

3.3.5 F1-Score

The F1 score is the harmonic mean of precision and recall, providing a balanced measure of the model's performance. By accounting for both precision and recall, the F1 score is suitable when there is an imbalance between positive and negative instances.

$$F1\ Score = 2\frac{(Precision * Recall)}{(Precision + Recall)}$$

3.4 Data preprocessing

3.4.1 Reverse CBAM

Our proposed custom attention mechanism, RCBAM (Reverse Convolutional Block Attention Module), introduces a novel approach to the traditional CBAM (Convolutional Block Attention Module) by reversing the order of attention operations and adding unique multi-scale feature extraction. Unlike the standard CBAM, which follows a sequential application of channel and spatial attention, our RCBAM first applies spatial attention followed by channel attention. Additionally, our method integrates multi-scale feature extraction to enhance the model's ability to capture important features at different scales.

Given:

- F: Input feature map
- $M_S(F)$: Spatial attention map
- $M_C(F)$: Channel attention map
- $\otimes$: Element-wise multiplication operator

The RCBAM module starts by inputting the feature map F. The feature map undergoes spatial attention first, generating F', followed by channel attention, resulting in F'' as shown in Fig. 2. The detailed process is as follows:

- **Spatial attention:** Compute the spatial attention map $M_S(F)$ and apply it into the feature map F using the following equations.
- Compute average-pooled and max-pooled feature maps:

$$F_{s,avg} = AvgPool(F)$$

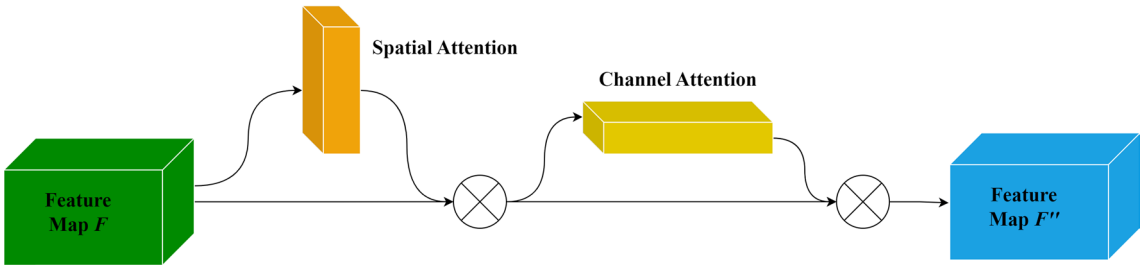


Fig. 2 The proposed RCBAM with spatial and channel attention used in this research

$$F_{s,max} = MaxPool\ (F)$$

- Concatenate the pooled feature maps along the channel axis:

$$F_{s,concat} = [F_{s,avg}, F_{s,max}]$$

- Apply convolution and sigmoid activation to obtain spatial attention map:

$$M_S(F) = \sigma(f^{7x7}(F_{s,concat}))$$

- Apply spatial attention map to the input feature map:

$$F\prime = M_S(F) \otimes F$$

- Channel attention: Compute the channel attention map $M_C(F')$ where F' is the output from the spatial attention step.
- Compute average-pooled and max-pooled feature maps from F'

$$F'_{c,avg} = GlobalAvgPool(F')$$

$$F'_{c,max} = GlobalmaxPool(F')$$

- Apply MLP to the pooled feature maps:

$$MLP(F'_{c,avg}) = W_1(W_0(F'_{c,avg}))$$

$$MLP(F'_{c,max}) = W_1(W_0(F'_{c,max}))$$

- A Combine the MLP outputs and apply sigmoid activation to obtain channel attention map:

$$M_c(F') = \sigma(MLP(F\prime_{c,avg}) + MLP(F\prime_{c,max}))$$

- Reshape the channel attention map and apply it to the spatially attended feature map F':

$$M_c(F') = \sigma(W_1(W_0(F\prime_{c,avg}) + W_1(W_0(F\prime_{c,max}))$$

$$F'' = M_C(F') \otimes F'$$

3.4.2 Hierarchical attention mechanism

To enhance the effectiveness of our attention mechanism, we employ a hierarchical attention mechanism with emphasis multi-scale feature extraction strategy. The input feature map is resized to different scales and then resized back to the original dimensions. This process captures features at various levels of granularity.

1. Input feature map: The original feature map F is resized to multiple scales and then concatenated:

   I. Original scale
   II. Half scale
   III. Quarter scale

- *Spatial attention*

  Spatial attention $M_S(F)$ is computed using average pooling and max pooling operations across the channel dimension, followed by a convolution operation with a sigmoid activation:

$$M_S(F) = \sigma(f^{7x7}\left[AvgPool(F) \parallel MaxPool(F)\right])$$

  where:
  σ: is the sigmoid function.
  f^{7x7}: is a convolutional operation with 7×7 filter size.
  $AvgPool(F)$ and $MaxPool(F)$ are the average and the max pooled feature maps along the channel axis.

- *Channel attention*

  Channel attention $M_C(F')$ is computed using global average pooling and global max pooling operations, followed by a multi-layer perceptron (MLP) with shared weights and a sigmoid activation:

$$M_C\left(F'\right) = \sigma(MLP(AvgPool\left(F'\right)) + MLP(MaxPool(F')))$$

  where:
  σ: is the sigmoid function.
  MLP: is a multi-layer perception with shared weights.

- *Fused attention*

  Global Average Pooling: Applied to the input feature map to generate a condensed representation.
  MLP: Generates dynamic attention weights to balance the contributions of multi-scale features.

$$W = \sigma(Dense_3(AvgPool(F))$$

  Combining these steps, the RCBAM process as shown in Figure 3 can be summarized as:

$$F' = M_S(F) \otimes F$$

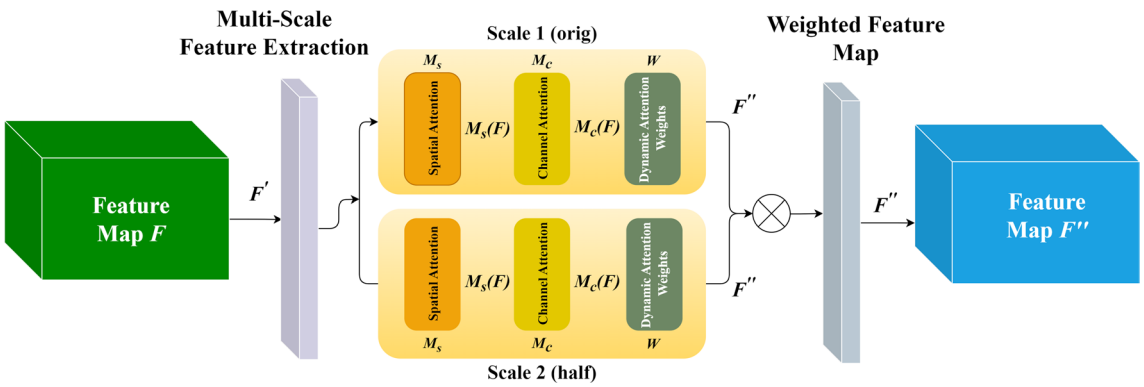


Fig. 3 The fusion of weighted feature maps with multiple scaling features extraction

$$F'' = M_C(F') \otimes F'$$

- **Adaptive attention mechanism**

I. *Multi-scale feature extraction*
Extract features at different scales: 1.0, 0.5, and 0.25.
Resize features to original input size for each scale.
Apply spatial and channel attention separately on each scale.

II. *Dynamic attention weights*
Extract features at different scales: 1.0, 0.5, and 0.25.
Resize features to original input size for each scale.
Apply spatial and channel attention separately on each scale.

3.4.3 MobilenetV2 base architecture with RCBAM

Regarding images, we employed a modified MobileNetV2 architecture with the RCBAM module to ensure robust feature extraction and mitigate overfitting as shown in Fig. 4. After training the model, Grad-CAM [28] was applied to the trained model to highlight regions of interest within the conjunctival images, providing visual explanations of the model's predictions. The highlighted regions from Grad-CAM were then used to extract color-based pixel percentage features, specifically focusing on the red, green, and blue components. These extracted features were saved into a CSV file. To create a multimodal dataset, we integrated this CSV file containing the pixel percentage features with additional textual data. By merging the image-derived features with the text data, we formed a comprehensive multimodal dataset that combines

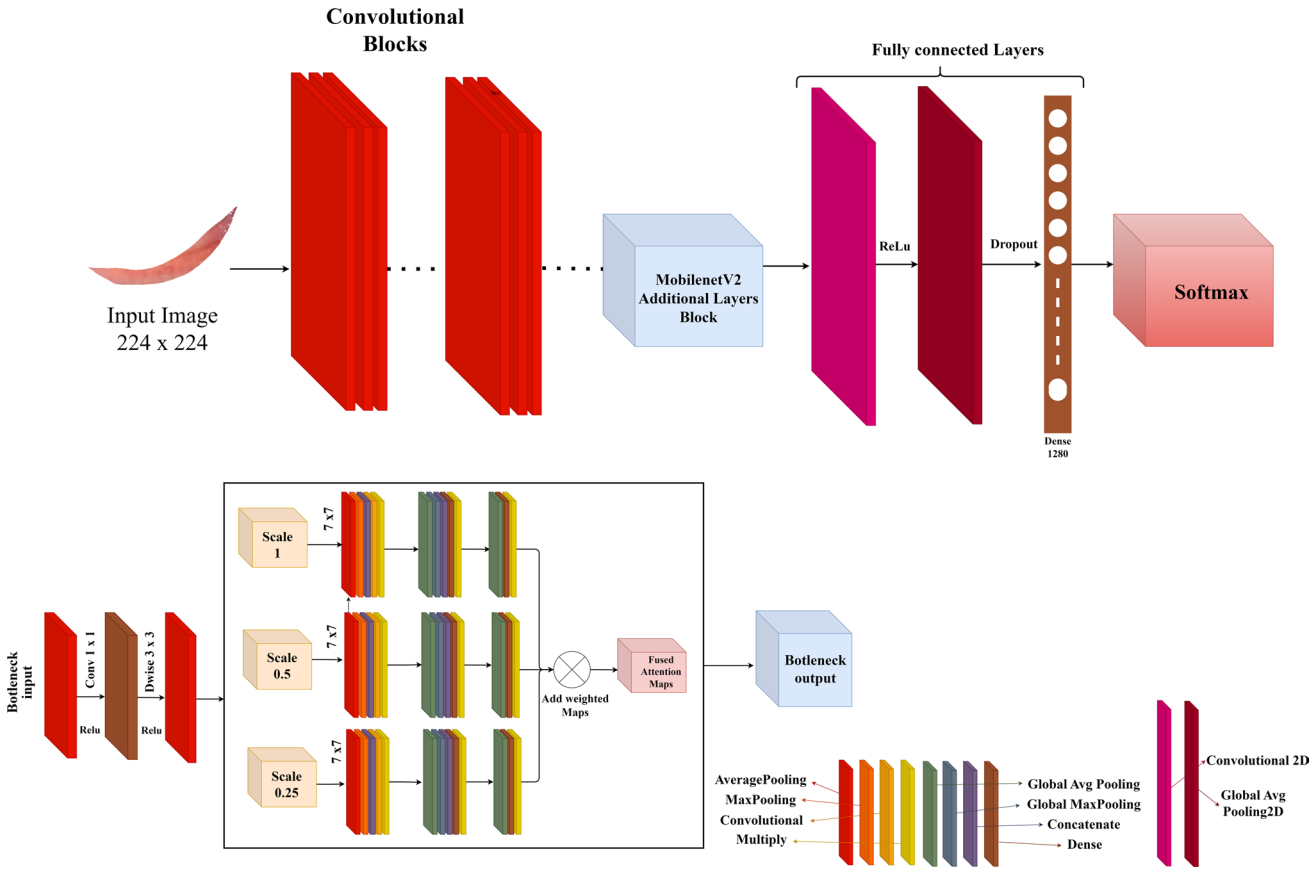


Fig. 4 The proposed model of RCBAM with MobilenetV2 base architecture

both visual and textual information. This approach allows us to leverage the strengths of both modalities, potentially enhancing the performance of subsequent analytical tasks such as classification and diagnosis.

3.4.4 Textual data

The project utilized EHR data, specifically focusing on the 'Gender' and 'Age(months)' as the target variable. For feature selection, both the odds ratio and chi-squared test were employed. The features were hand-crafted as discussed in the previous steps. A medical professional reviewed the odds ratio and chi-squared test results, incorporating their medical and statistical expertise to select the most relevant features. These top features, identified based on their potential to classify anemia, will be used in conjunction with image-extracted features in our model to develop a non-invasive diagnostic technique.

I. Exploratory data analysis

We need to explore/clean/filter it before processing it for ML It involves adding/deleting few columns or rows, joining some other data, and handling qualitative variables like dates. Where there are 710 instances in the dataset with 10 columns. The dataset has 710 records with 5 features as show in Fig. 5, taking up 80KB of memory. It's clean with no missing values. Data types: 6 object, others float64. A secure copy was made to prevent changes. Labels were assigned for clarity: '0'/'1' for 'Non-Anemic'/'Anemic' and '0'/'1' for 'Male'/'Female' to aid research paper analysis.

Addressing class imbalance is crucial in building robust predictive models. Imbalanced classes can skew model performance, leading to biased predictions favoring the majority class while neglecting the minority class. In our dataset, the prevalence of non-anemic cases outweighs anemic ones, posing a challenge for accurate modeling. Without proper handling of this imbalance, predictive algorithms are prone to overfitting and erroneous outcomes, as they may incorrectly assume the dominance of the majority class. Therefore, it is imperative to rebalance the classes in the data to ensure unbiased and reliable predictions. In this dataset, female exhibit a 57% anemia prevalence, while male show a significantly higher rate of 62%. This disparity suggests a greater likelihood of females experiencing anemia. Additionally, the male population surpasses females by 5%. These findings underscore the importance of gender-specific health interventions and warrant further investigation into underlying factors.

The odds ratio of anemia between males and females in the dataset is approximately 1.23, indicating that males have 23% higher odds of anemia compared to females. A chi-square test was conducted to examine the association between age and gender. The results yielded a chi-square statistic of 7.69 with a p-value of 0.0529. The p-value, being slightly above the conventional significance level of 0.05, suggests a marginal association between age and gender. Although this indicates a potential slight relationship. However, if a more lenient significance threshold (e.g., 0.1) is considered, the p-value of 0.0529 would suggest a weak but notable association between age and gender. Despite the marginal nature of this association, our medical professional reviewed the odds ratio and chi-squared test results, incorporating their medical and statistical expertise. Based on their experience and knowledge of medical health, they recommended including age as a feature due to its potential relevance in classifying anemia.

It's observed that the maximum value of Hb concentration within the studied population reaches 13 units. Furthermore, our analysis indicates a notable left skewness in the distribution of Hb levels, as evidenced by a more extended tail on the left side of its peak. This skewness suggests that a considerable portion of the population exhibits lower Hb concentrations relative to the peak value. Lower Hb levels are indicative of anemia, a condition characterized by reduced oxygen-carrying capacity in the blood. The normal range for Hb concentration typically falls within 13.2 to 16.6 grams per deciliter (g/dL) for men and 11.6 to 15 g/dL for women. Specifically, Hb levels below 13.2 g/dL are considered low for men, while levels below 11.6 g/dL are deemed low for women.

The odds ratio, a key measure of association between variables, reveals the likelihood of anemia occurrence based on gender. In our logistic regression, with 5 iterations, the odds ratio for gender is 3.41, indicating females have 3.41 times higher odds of anemia than males. This underscores a gender discrepancy in anemia susceptibility, with females at elevated risk.

3.5 Multimodal data fusion

The combination of modalities involves integrating extracted features from EHRs and Conjunctival Images. This integration seeks to blend the complementary information from these diverse sources, forming a cohesive representation

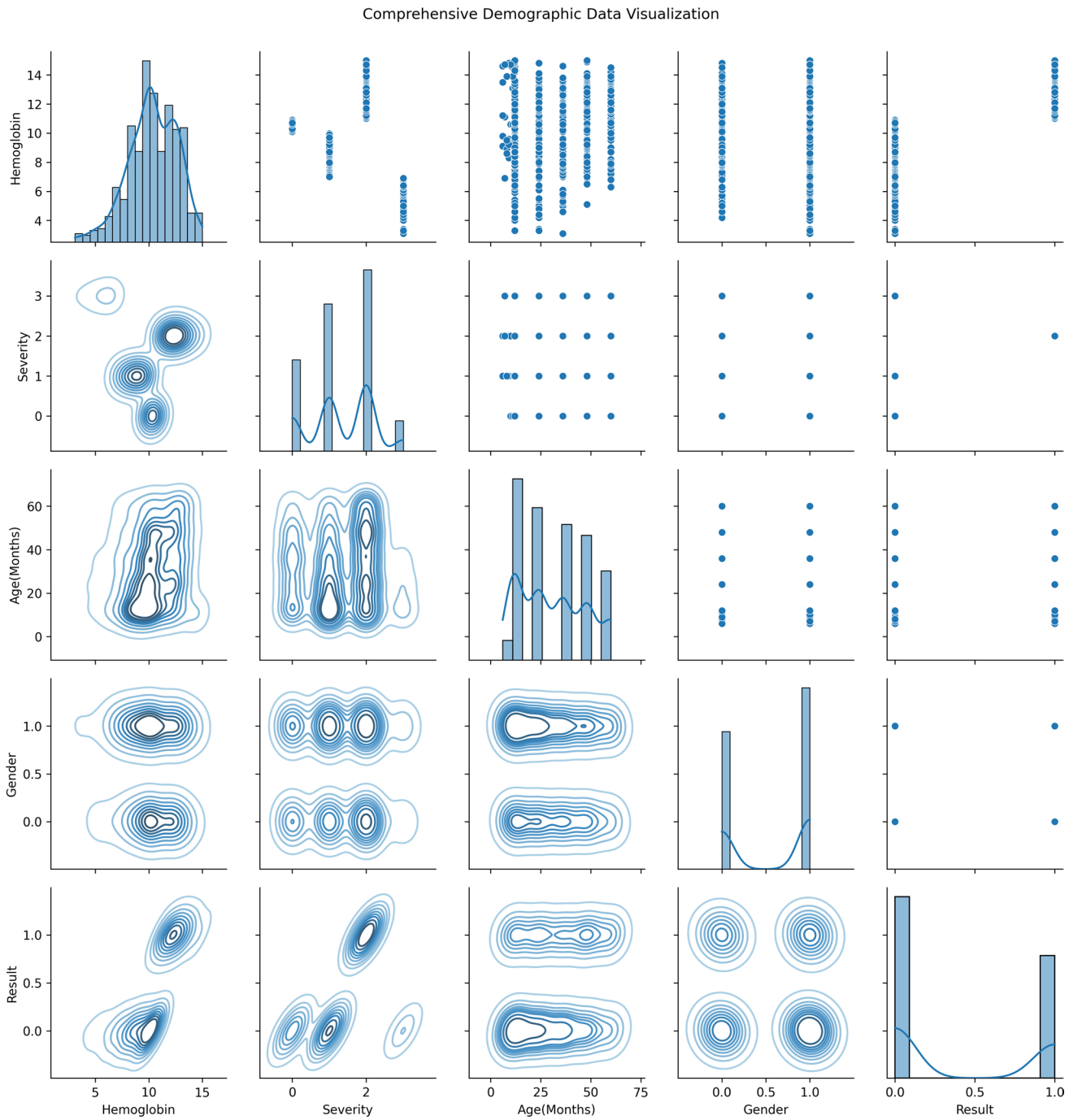


Fig. 5 The demographic Information of the patients based on textual dataset

that encompasses the overarching traits of anemia. The choice of fusion algorithm is crucial and depends on the characteristics of the data and the desired levels of interpretability. In this project, the fusion of EHR data with Conjunctival Images extracted features data is prioritized to enable the development of a comprehensive anemia detection model as shown in Fig. 6

The advantages of employing a multimodal fusion technique are manifold. Firstly, it offers a non-invasive approach, circumventing the need for invasive procedures such as blood tests. This enhances patient-friendliness and accessibility to screening and diagnostic processes. Moreover, by amalgamating information from multiple modalities, the technique enhances accuracy and reliability in anemia detection when compared to single-modality methods. This heightened accuracy contributes to early detection capabilities, potentially enabling timely intervention and treatment, thus mitigating

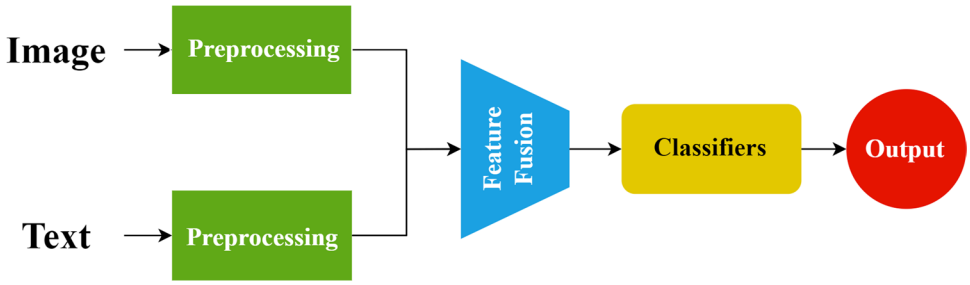


Fig. 6 The basic flow of multi-modal features fusion of both data modalities

the progression and severity of anemia-related complications. Additionally, the multimodal fusion technique facilitates personalized diagnosis by considering individual patient characteristics and risk factors, leading to tailored treatment plans and better patient outcomes.

The research faces several limitations that merit consideration. Firstly, the dataset used exhibits a notable class imbalance, skewing towards non-anemic cases, which can bias model predictions and potentially overlook critical patterns in anemia detection. While mitigation strategies like oversampling and SMOTE were applied, their efficacy in capturing the full complexity of anemia classification remains uncertain. Secondly, the study's findings may lack generalizability beyond its specific dataset of conjunctival images and EHRs, as variations in demographics, environmental factors, and image quality could impact model performance in broader clinical contexts. Thirdly, the method of feature selection, guided by odds ratios and chi-squared tests, may oversimplify complex interactions and latent variables crucial for comprehensive anemia diagnosis. Moreover, while multiple classifiers were evaluated using standard metrics, the adequacy of these metrics in clinical practice, especially in managing false positives and false negatives, requires further exploration. Additionally, the computational demands of techniques like GridsearchCV (Grid Search with Cross-Validation) for hyperparameter tuning pose scalability challenges without sufficient computational resources. Finally, while Grad-CAM provides visual insights into model decisions based on conjunctival images, interpreting these features in clinical decision-making remains a significant hurdle. Addressing these limitations is crucial for refining the proposed multimodal approach and enhancing its applicability and reliability in real-world anemia diagnosis scenarios.

4 Results and discussion

This research aimed to develop an effective anemia detection model by leveraging a multimodal fusion approach that combines EHRs data with image-derived features. The model's robust performance underscores its potential for clinical application in identifying individuals with and without anemia.

4.1 Results of proposed model on image and textual dataset

The Table 2 presents a comparison of performance metrics for various machine learning models applied to textual and image datasets. For the textual dataset, the Random Forest (RF) model achieved the highest accuracy at 96.33%, whereas the Support Vector Machine (SVM) exhibited the lowest accuracy at 75.06%, the Improved RCBAM-based MobileNetV2

Table 2 The results of the proposed model on image and textual dataset

Models	Dataset	Accuracy	AUC	Precision	Recall	F1 Score
LR	Textual Dataset	94.36%	92.4%	93.19%	94.63%	94.19%
DT		93.12%	94.21%	90.60%	94.92%	95.01%
RF		96.33%	96.0%	94.36%	95.09%	94.98%
SVM		75.06%	77.65%	82.62%	80.19%	79.67%
NB		91.16%	88.0%	90.65%	92.97%	94.31%
KNN		94.03%	95.23%	92.35%	94.25%	91.67%
RCBAM based MobilenetV2	Image Dataset	82.45%	79.0%	86.71%	87.34%	86.48%
CBAM based MobilenetV2		79.43	80.35	78.76	79.67	79.27
MobilenetV2		75.11	76.00	75.23	74.03	75.12

Table 3 The results of the proposed model using feature fusion approach

Models	Dataset	Accuracy (%)	AUC (%)	Precision (%)	Recall (%)	F1 Score (%)
RCBAM based MobilnetV2	Image Dataset + Textual data	80.51	82.78	77.78	78.98	74.35
CBAM based MobilenetV2		76.48	78.65	77.98	78.32	79.01
MobilenetV2		73.12	75.53	76.07	75.84	75.25

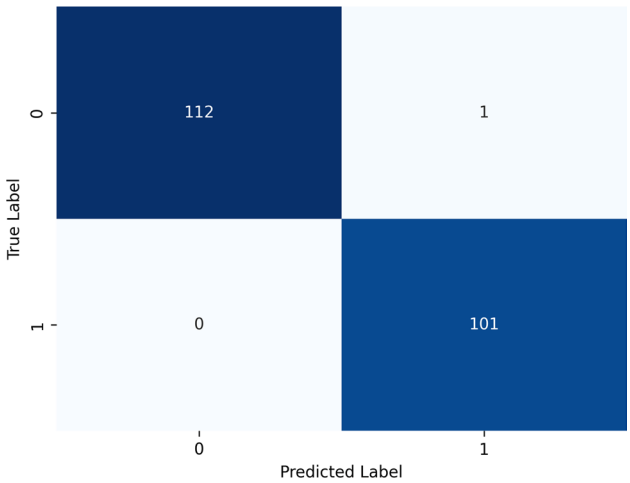


Fig. 7 The confusion matrix of the proposed model for anemia detection

attained an accuracy of 83.45%. The metrics assessed include Accuracy, Precision, Recall, and F1 Score, underscoring the effectiveness of each model across different data types.

4.2 Results of the proposed model using feature fusion approach

Features from both the image and textual data were utilized and processed through a model, with SelectKBest used for feature selection to optimize training and testing. However, the RCBAM based MobileNetV2 model did not perform optimally on the multimodal fusion, achieving an accuracy of 80.51%, AUC of 82.78%, precision of 77.78%, recall of 78.98%, and an F1 score of 74.35%, as shown in the Table 3.

4.3 Results of the proposed model using handcrafted multimodal approach

The handcrafted features were meticulously extracted based on the model's requirements to optimize performance across various metrics. These features were developed in consultation with medical professionals to ensure the model remains entirely non-invasive. Examining the confusion matrix Fig. 7, we found that the model demonstrated high recall in correctly identifying individuals without anemia, achieving a rate of 94%. However, there were instances of false positives in predicting anemia, leading to an 83% precision for anemic cases. While the model excelled in capturing all instances of anemia (recall of 1.00), the occurrence of false positives suggests room for refinement, particularly in minimizing erroneous classifications of anemia.

The analysis of the Receiver Operating Characteristic (ROC) curve further supported the model's efficacy, with an area under the curve (AUC) of 1.00 and Matthew's Correlation Coefficient 0.88 indicating excellent discriminatory power in distinguishing between anemic and non-anemic individuals. This suggests that the model can effectively balance sensitivity and specificity in detecting anemia as shown in Fig. 8.

Feature importance analysis revealed that Age (Months) Gender and Image extracted features pixel percentage significantly contributed to the model's predictions in Fig. 9. These findings underscore the clinical relevance of incorporating both clinical data and visual information in anemia detection, highlighting the importance of multimodal fusion techniques.

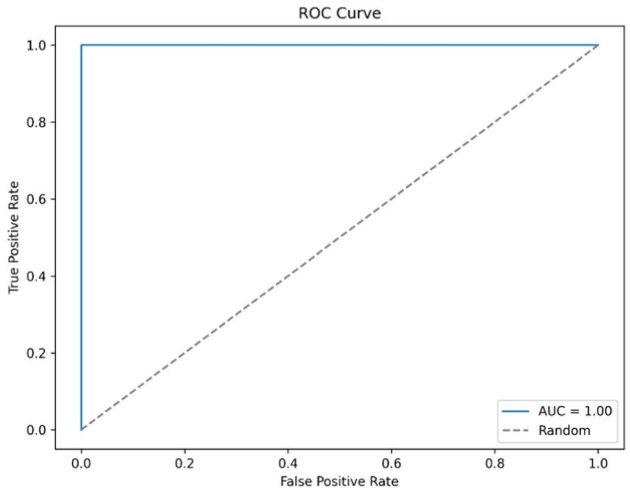


Fig. 8 The ROC curve of the proposed model for anemia detection

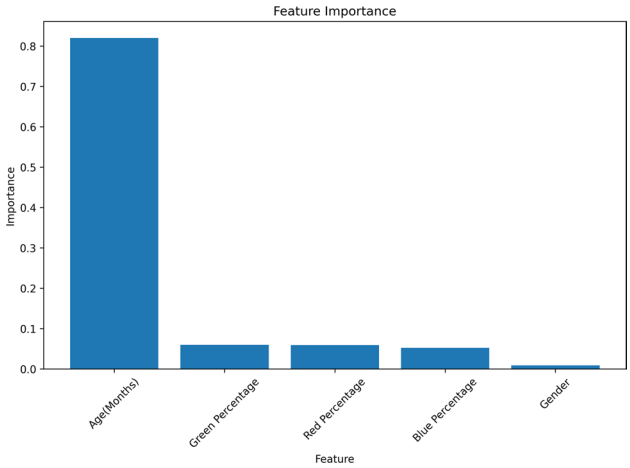


Fig. 9 The overview of feature importance for the anemia dataset

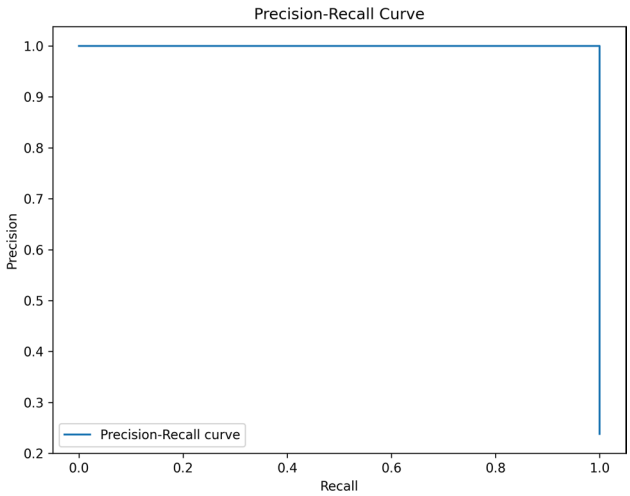


Fig. 10 The precision-recall curve of the proposed model

Despite its strengths, including high precision and recall in identifying individuals without anemia and the ability to leverage diverse data sources, the model still has areas for improvement as shown in Fig. 10. Minimizing false positives in anemic cases remains a priority, as these errors can have significant clinical implications.

As in Fig. 11 calibration curve visualizes the relationship between the predicted probabilities from our model and the actual outcomes. Ideally, a perfectly calibrated model would have its predictions perfectly aligned with the actual outcomes, represented by the diagonal line in the graph. As you can see, the blue line representing our model's performance

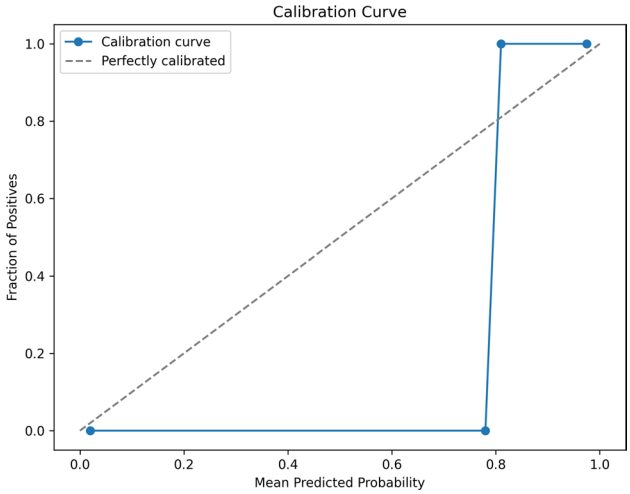


Fig. 11 The calibration curve of the proposed model

closely follows the diagonal, indicating a strong correlation between predicted probabilities and actual outcomes. This suggests that our model can accurately estimate the likelihood of an event occurring.

Notably, our model achieves an impressive 95% accuracy, further solidifying its effectiveness in predicting the desired outcome. The model's performance is also evidenced by its precision and recall metrics, with a precision of 96.0%, a recall of 95.93.0%, an F1-score of 94.0%, and an AUC of 95.0%. Additionally, enhancing the model's interpretability and clinical relevance through domain expertise consultation could further optimize its utility in real-world healthcare settings.

Test Data 1: In this case, the individual is female (Gender = 1) with a Hb level of 11, which is below the threshold of 13.45. The model correctly predicts anemia for this case by following the RF path based on gender and Hb levels.

Test Data 2: Similarly, in the second test case, the individual has a Hb level of 11, which falls below the threshold of 13.45. The model effectively navigates the RF, taking into account both gender and Hb levels, and successfully identifies anemia due to the aligned criteria.

These test cases highlight the model's ability to accurately predict anemia based on the provided features, reinforcing the reliability of the RF model in our research for anemia diagnosis.

4.4 Ablation study

In this ablation study, we investigate the efficacy of a fusion model integrating various classifiers—SVM (SVM) with best parameters C 0.1 and a linear kernel, DT (DT) with gini criterion and max depth 1, LR (LR) best parameter C 0.01 and solver newton-cg, k-NN with parameters 19 number of n-neighbors with auto algorithm and RF (RF) with best parameters max depth 1 and n_estimators 50—to enhance classification accuracy. Each fusion model configuration is denoted by the presence (√) or absence (×) of individual classifiers. For instance, Fusion SVM incorporates only SVM while excluding DT, LR, and RF as shown in the Table 4. By systematically varying the composition of classifiers within the fusion model, we assess their respective contributions to overall classification performance. The table displays the accuracy achieved by each fusion model, providing insights into the impact of individual classifiers on the fusion model's effectiveness. This ablation analysis offers valuable insights into optimizing classifier combinations for improved classification outcomes, crucial for enhancing the robustness and efficiency of ML systems.

Table 4 The results of the ablation conducted using different methods with feature fusion of dataset

Methods	Fusion SVM	Fusion DT	Fusion LR	Fusion KNN	Fusion RF	Accuracy %	AUC %	Precision %	Recall %	F1 Score %
SVM	√	×	×	×	×	93.7	80.0	80.0	79.89	71.23
DT	×	√	×	×	×	90.4	90.0	88.0	88.0	88.4
LR	×	×	√	×	×	88.3	79.0	81.0	80.07	82.3
KNN	×	×	×	√	×	86.01	88.9	82.14	82.0	87.4
RF	×	×	×	×	√	95.0	95.0	96.0	95.93	94.0

Table 5 The results of the handcrafted features on real-time data

Image_Details	Time	Date	Gender	Age (Months)	Results
IMG20240416246611	19:00	2024-04-16	Male	36	Anemic
IMG20240418246625	20:38	2024-04-18	Male	44	Non-Anemic
IMG20240420246632	22:13	2024-04-20	Female	35	Anemic
IMG20240422246633	17:15	2024-04-22	Male	40	Anemic
IMG20240427246645	16:17	2024-04-27	Female	48	Non-Anemic

4.5 In comparison of the proposed handcrafted feature extraction

The thorough assessment of different models for detecting anemia, as described in the document, illustrates a wide range of methodologies and their respective outcomes. These models underwent rigorous testing using real-time data collected during a clinical trial held at a hospital in Pakistan, detailed in Table 5. The trial was conducted under the supervision of a healthcare expert—a doctor with three years of practical experience and an MBBS degree from Shandon First Medical University, China. The findings from this trial were meticulously validated and authenticated under the oversight of the mentioned healthcare professional.

In comparison to previously conducted studies [20, 24, 25] in the domain of medical image analysis for conjunctiva as shown in Table 6, certain gaps were identified. References [24] utilized diverse datasets, potentially incurring higher computational costs, and mainly focused on a single-model approach, hindering comparisons between methodologies. Roychowdhury [25] explored both conjunctival and palpable tongue images, introducing a dual-dimensional perspective.

The proposed research addresses these gaps by employing a targeted dataset consisting of 104 patient images of the conjunctiva, thereby reducing computational expenses and ensuring a more focused analysis. We introduce a novel color-based pixel percentage extraction technique, complemented by the handcrafted feature extraction, for multimodal data fusion. Notably, our use of the RF classifier resulted in a remarkable 95% accuracy, surpassing the achievements of prior studies. Moreover, our study maintains alignment with existing literature by incorporating conjunctival images while introducing a new dimension through its specific patient population. This research not only contributes to the field of ocular tissue analysis but also effectively bridges critical gaps observed in the referenced studies.

5 Conclusion

This research introduces a novel non-invasive technique for automated anemia detection using multimodal data fusion by combining EHRs and conjunctival images. First, both the datasets are preprocessed with different methods and the best features are selected. After that, these features are combined to form a multimodal dataset. The standard architecture of the MobileNetv2 is modified by adding a novel RCBAM block for better feature extraction. Different experiments are performed on textual and image datasets individually and after combining. The ablation experiments are also conducted to know the effectiveness of the proposed model. The results show that the proposed model achieves accuracy of 95%. The proposed methodology opens some promising directions for future research. The real-time anemia detection systems can be introduced by using other information such as genetic information, and multiple imaging modalities. The anemia detection studies can be integrated in wearable devices to keep monitoring of anemia.

Table 6 The comparative analysis of the proposed model with previous methods

Ref	Dataset	Preprocessing	Algorithms used	Contribution
[20]	19 conjunctivas of the eye images	Image segmentation	RGB thresholding algorithm	The RGB thresholding algorithms used attained an accuracy of 78.90%
[24]	99 images of the conjunctiva	Image augmentation and image segmentation	Aartificial Neural Network (ANN)	An accuracy of 97.00% was attained by the ANN
[25]	27 conjunctiva of the eyes and 56 palpable tongue images	Image segmentation	SVM, LR, DT, decision, ANN, Poisson regression, and k-NN	86% accuracy for eyes conjunctiva and 98.2% accuracy for palpable tongue with the DT algorithms
Proposed Model	710 Patients' Images of the conjunctiva and Health record data	Color-based pixel percentage and handcrafted features extraction technique	RF	95% Accuracy Attained by the multimodal data fusion

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Author contributions MR conducted the experiments and wrote the manuscript; MU proposed the initial idea and revised the manuscript; GA performed the formal analysis and software; all the authors read and approved the final manuscript.

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Data availability The data used in this study is openly available from Mendeley repository at (https://data.mendeley.com/datasets/m53vz6b7fx/1) or the author MR can be contacted for the availability of the data. The authors accessed this data under the terms of the Mendeley license, and interested researchers can freely access it for non-commercial purposes.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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